TECHNICAL CASE STUDY / NOVEMBER 2026

Megalophobia

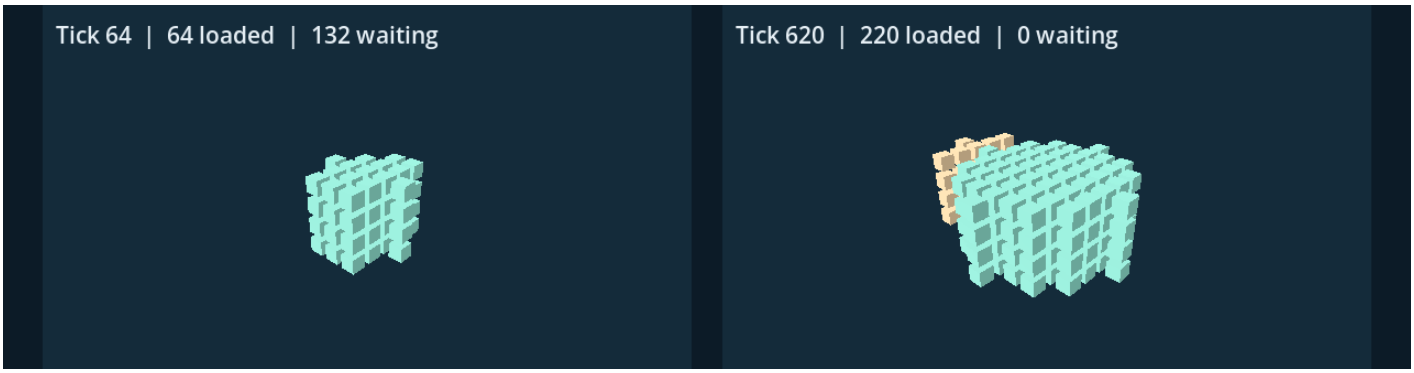
Engineering a procedural voxel world

https://github.com/MindScrape/Megalophobia

An independent voxel prototype in Godot 4.6 and GDScript. The engineering focus is deciding which work to do, reducing the amount of work, and keeping player edits consistent as background tasks finish.

1 Stream terrain with bounded work

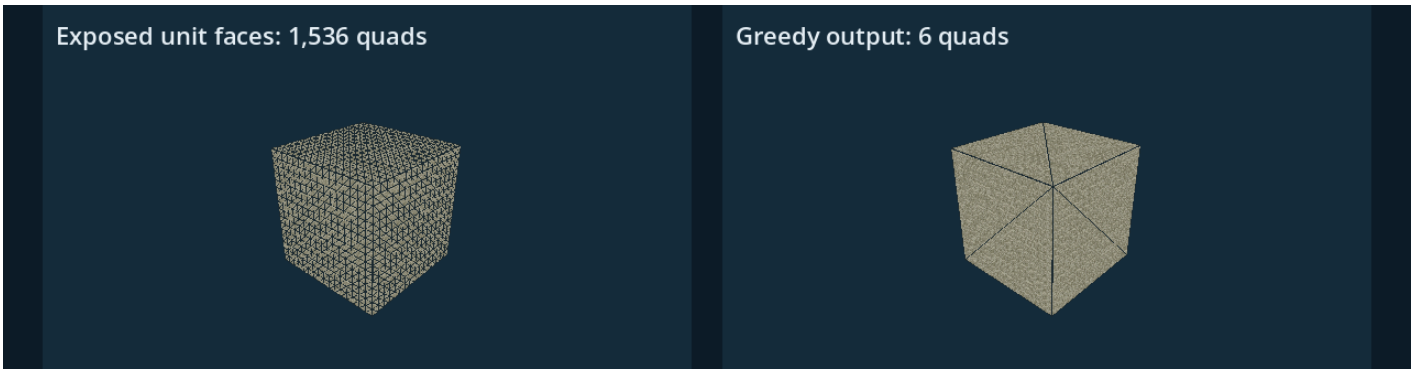
The world loads nearby 16 x 16 x 16 chunks first. Separate frame budgets spread loading and unloading over time; a wider unload radius reduces churn at the boundary. Background rebuilds have an eight-task limit, while player edits receive priority.



Scheduler fixture: 620 manual ticks, radius 4, and an eight-chunk move. The observed limits were one spawn and two unload attempts per tick. Terrain generation was excluded.

2 Reduce geometry before rendering

Greedy meshing merges adjacent faces with the same block type and orientation. The atlas shader repeats block textures across those larger rectangles. In one procedural chunk, 3,234 exposed unit faces became 1,528 merged quads. Surface shape determines how much can merge.



The isolated solid-chunk comparison uses the same outer surface: 1,536 unit quads versus six merged quads. This is a geometry reduction, not an FPS speedup.

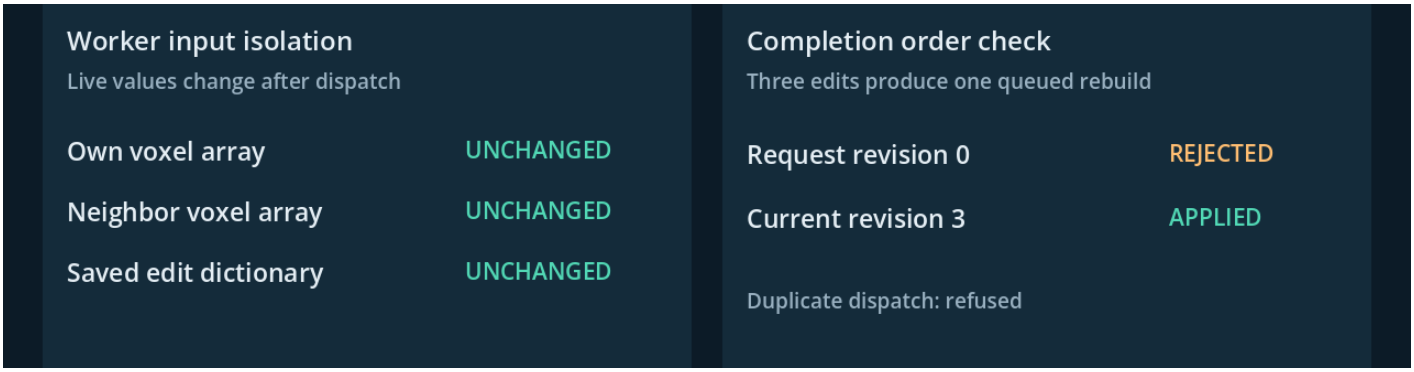
Source hashes, raw results, and reproduction commands accompany this study in docs/technical-case-study.md. Measurements cover controlled fixtures rather than gameplay frame time.

Keep edits consistent

The prototype separates worker input isolation, stale-result rejection, and change storage. Each protection has a focused, repeatable check.

3 Keep asynchronous edits consistent

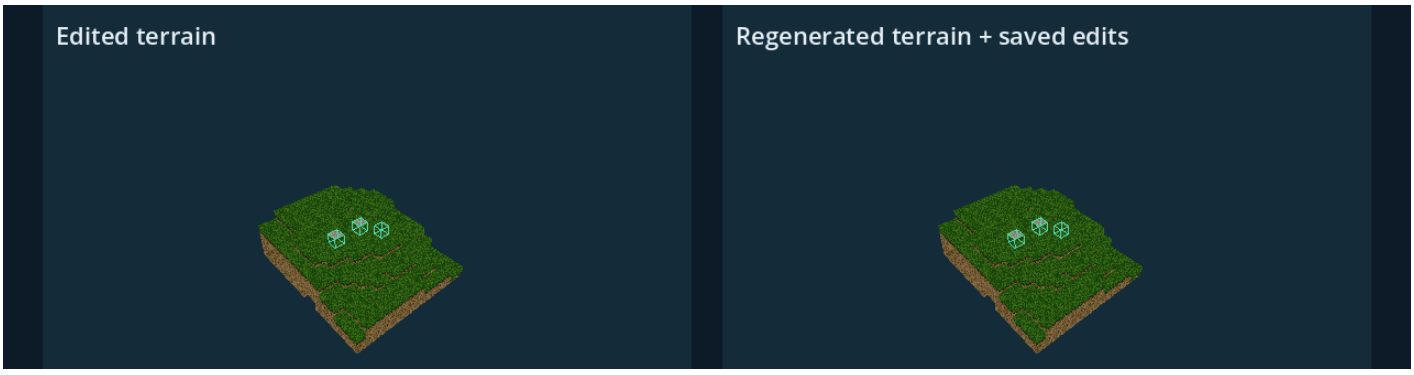
A request now explicitly copies its own voxels, neighboring voxels, and saved edit data before entering the worker pool. A per-chunk guard rejects duplicate dispatch, and revision checks discard stale mesh results. Repeated edits share one pending rebuild entry.



A semaphore-controlled worker regression passes all five checks after the copy fix. Separately, injected completions show revision 0 rejected and revision 3 applied after three edits.

4 Persist changes to a reproducible world

Fixed seeds regenerate the base terrain. Storage records edited voxel indices and replacement block IDs, then replays them on load. In a three-edit fixture, the saved dictionary occupied 60 bytes versus 16,396 bytes for the full chunk in the same serialization format. Reconstructed arrays matched exactly.



The storage helpers were exercised directly with temporary files. Normal debug gameplay disables disk saves; release-mode save and failure recovery are separate validation work.

Greedy CPU sample: 15.49 ms median, 15.72 ms p95; 30 runs after 3 warmups on Apple M4, Godot 4.6.1 debug. Axis passes only, excluding generation, upload, collision, and rendering.

Next: empty-mesh handling, reliable unload retries, boundary-face ownership, complete task cleanup, and failed-save recovery. These are tracked with verification criteria in the engineering backlog.